

System Workflow Document

This document outlines the core workflows within the **ZAIZEN Financial Intelligence** platform.

1. User Authentication & Authorization Workflow

The application uses a JWT-based authentication system with Role-Based Access Control (RBAC).

```
sequenceDiagram
    participant User
    participant Frontend
    participant AuthAPI
    participant Database

    User->>Frontend: Enter Credentials
    Frontend->>AuthAPI: POST /api/auth/login
    AuthAPI->>Database: Verify User & Password
    Database-->>AuthAPI: User found, Hash matches
    AuthAPI->>AuthAPI: Generate JWT (User ID, Roles, Expiry)
    AuthAPI-->>Frontend: HTTP 200 (JWT + Profile)
    Frontend->>Frontend: Store JWT in LocalStorage
    Frontend->>User: Redirect to Dashboard
```

2. Financial Entry & Audit Workflow (WORM)

For financial integrity, journal entries follow a strict "Write Once Read Many" (WORM) pattern through an immutable audit trail.

```
graph TD
    A[User creates Journal Entry] --> B{Validates Balance?}
    B -- No --> C[Error: Debits != Credits]
    B -- Yes --> D[Store in JournalEntries table]
    D --> E[Log to Immutable Audit Service]
    E --> F[Entry status: Unvalidated]
    F --> G[Authorized User validates Entry]
    G --> H[Entry status: Validated]
    H --> I[Update Financial Reports]
```

3. Expense Management Workflow

Handles expense reporting, receipt uploads, and ledger posting.

```
sequenceDiagram
    participant Staff
    participant Manager
    participant API
    participant Storage

    Staff->>API: Create Expense Report
    Staff->>API: Add Lines (Amount, Category, Project)
    Staff->>API: Upload Receipt Image
    API->>Storage: Save to wwwroot/uploads
    Staff->>API: Submit for Review
    Manager->>API: Approve & Post to Ledger
    API->>API: Generate Journal Entry (44xxx to 6xxx)
```

4. ERP & Commercial Workflow (Procurement to Warehouse)

```
graph LR
    P[Purchase Order] --> S[Supplier Approval]
    S --> GR[Goods Receipt]
    GR --> W[Warehouse Update]
    W --> I[Inventory Movement Log]
    I --> J[Auto-Journal: Inventory Increase]
```

5. Security Guardrails

Component	Logic	Purpose
JWT Bearer	RSA/HMAC Signature	Verifies identity of the requestor
Membership Filter	UserCompanies check	Prevents users from accessing data of companies they don't belong to

RBAC	<code>RequirePermission</code>	Restricts features (e.g. only Accountants can validate entries)
WORM Audit	<code>ImmutableAuditService</code>	Ensures no one can delete or modify past transaction logs